

CS432/532: Final Project Report

Project Title: Spotify Data Analysis

Team Member(s):

Bhondve Aakanksha Prashant
Gadikar Akanksha Uday
Mukherjee Prerana

I. PROBLEM

This project will analyse the Spotify songs, artists, genres and Music patterns/trends of the songs. We have used public datasets made available from Kaggle to examine the characteristics of the song from different countries, regions and languages.

The project comprises four analyses that provide information on tracks, artists, genres audio features and rank of the Spotify weekly top songs from the dataset.

1. Analysis I

Analyse artist names and track names on the basis of streams and colabs in the year 2022.

2. Analysis II

Analyse the previous and current week's rank of track in the year 2022

3. Analysis III

Analyse the tracks based on audio features and streams for country Argentina

4. Analysis IV

Analyse track names based on popularity on charts for each artist genre for year 2021 to 2022.

II. SOFTWARE DESIGN AND IMPLEMENTATION

A. Software Design and NoSQL-Database and Tools Used

In this project, MongoDB used as NoSQL database, opting for its ease of use and capacity to manage and analyse semi-structured data effectively. MongoDB Atlas served as our cloud-based database solution, offering a web interface that facilitated seamless interaction with our data. This setup granted us the flexibility to connect to the database at any given moment and securely store all our valuable data.

The Gui developed has allowed us to examine collections and the primary outcomes achieved using Python scripts. These scripts helped in filtering and sorting from MongoDB . The

pymongo library from Python was used to connect to and interact with the database.

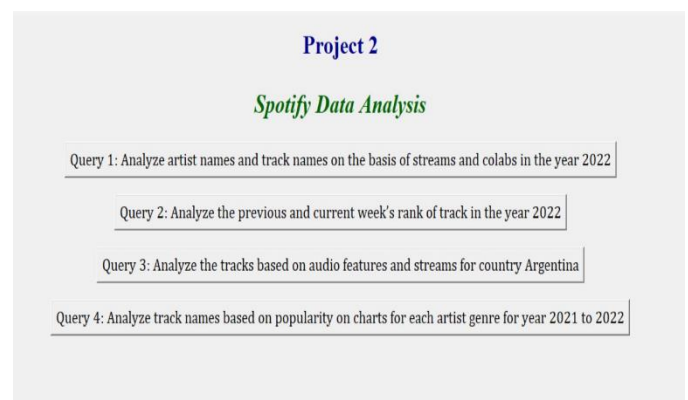
B. Parts that you have implemented

Implementations performed for this project are as follows:

The tables that have been visualized are:

- Visualization to get artist and track name streaming for year 2022 .
- Analyzing the ranks (previous and recent) for tracks streamed for year 2022.
- Analyze audio features of tracks streamed from Argentina.
- Evaluate each genre's songs, present on Spotify charts for maximum number of weeks for duration of 2021 - 2022.

III. PROJECT OUTCOME



GUI Front page

Analysis I

Analyze artist names and track names on the basis of streams and colabs in the year 2022. The first query is based on finding

the top 20 trends with the artists names and track names that were released in the year 2022. It is based on their number of streams which means if the stream value is highest then considered in the top list. We have taken another attribute as colab with value “0”. Here, colab is the column in which if the value is “0” then a single artist has made the track and if it is “1”, then multiple artists together have done the track. We have considered tracks made by a single artist in this query. In the Query 1 Result below, we can see top 20 tracks having maximum number of streams from the year 2022 along with their artist name.

Query 1 Result:

Sr. No.	Track Name	Artist Name	Number of Streams
1	As It Was	Harry Styles	1760962147
2	First Class	Jack Harlow	716610859
3	Titi Me Preguntó	Bad Bunny	634729651
4	Moscow Mule	Bad Bunny	581858025
5	Una Noche en Medellín	Cris Mj	555742632
6	PROVENZA	KAROL G	508844213
7	Light Switch	Charlie Puth	478987563
8	About Damn Time	Lizzo	463033925
9	Efecto	Bad Bunny	462681191
10	Boyfriend	Dove Cameron	433475301

Query 1 Result:

Sr. No.	Track Name	Artist Name	Number of Streams
11	Glimpse of Us	Joji	423476333
12	Late Night Talking	Harry Styles	370070833
13	Después de la Playa	Bad Bunny	349616901
14	Envolver	Anitta	341850122
15	Fuera del mercado	Danny Ocean	338366370
16	she's all i wanna be	Tate McRae	335722525
17	Tacones Rojos	Sebastian Yatra	329165854
18	Un Ratito	Bad Bunny	306591552
19	Fingers Crossed	Lauren Spencer-Smith	296877232
20	Yo No Soy Celoso	Bad Bunny	260714024

Query 1 Result

Analysis II

Analyze the previous and current week’s rank of track in the year 2022. In this query, we compare the current rank in the year 2022 previous week's top rank of a track, and show the corresponding artist name, track name and respective ranks.

Query 2 Result:

Sr. No.	Track Name	Artist Name	Current Rank	Previous Rank
1	W / X / Y	Tani Yuuki	1	1
2	A+Γ=BH/I	Galena, Azis	1	1
3	∞ LOVE (feat. Guè)	Marracash, Guè	1	1
4	Pillow Talk	MC 張天賦	1	1
5	Pohjola	Olli Halonen	1	1
6	Rozmazana kreska	Kinny Zimmer	1	1
7	Red Scarf - "Till We Meet Again	WeiBird	1	1
8	Chinchilla	SNIK, FLY LO, Mike G	1	1
9	Rockin' Around The Christmas	Brenda Lee	34	1
10	Tabara	YNY Sebi, Petre Stefan, Aron, D	1	1

Query 2 Result:

Sr. No.	Track Name	Artist Name	Current Rank	Previous Rank
11	All I Want for Christmas Is You	Mariah Carey	40	1
12	Stiny	Viktor Sheen	1	1
13	Ánh Sao Và Bầu Trời	T.R.I	1	1
14	Dvel ég í draumahöll	Hafdis Huld	1	1
15	Dear My Friend,	Keung To	1	1
16	Salade Coco	ElGrandeToto	1	1
17	Etnaset	Muslim - مسلم	1	1
18	Snowman	Sia	45	1
19	PAPARAZZI	UZI	1	1
20	Další rok	Smack	1	1

Query 2 Result

Analysis III

Analyze the tracks based on audio features and streams for country Argentina. In third analysis , track names which are highly streamed on Spotify charts and have maximum value for features danceability , energy , loudness , valence and speechlines and liveness . Analysis can be used for classifying the tracks as happy songs , high energy songs or sad songs based on above feature values .

Query 3 Result:

Sr. No.	Track Name	Max Streams	Max Danceability	Max Energy	Max Loudness	Max Valence	Max Speechlines	Max Liveness
1	Quevedo: Bzrp Music	6065170	0.621	0.782	-5.54799999999	0.55	0.0440000000000	0.23
2	Nicky Jam: Bzrp Music	4704707	0.627	0.849000000000	-3.167	0.818	0.153	0.145
3	Plan A	4548868	0.583	0.834000000000	-4.875	0.557	0.0444	0.0658
4	Bar	4416469	0.523	0.691	-3.802	0.677	0.0833	0.327
5	Todo De Ti	4372773	0.78	0.719000000000	-3.605	0.342	0.0506	0.0932
6	Militeense	4240815	0.85	0.541	-4.522	0.916	0.0674	0.286000000000
7	Entre Nosotras	3998977	0.729000000000	0.513	-6.345	0.789	0.0455	0.136999999999
8	2-50 Remix	3774689	0.819000000000	0.812	-4.425	0.844000000000	0.0478	0.234
9	Qué Más Pues?	3613767	0.903	0.819000000000	-3.964	0.768	0.106	0.242
10	L-Gante: Bzrp Music	3610015	0.552999999999	0.819000000000	-3.083	0.760999999999	0.257	0.0915

Query 3 Result

Analysis IV

Analyze track names based on popularity on charts for each artist genre for year 2021 to 2022. In this query ,analyzing each artist genre’s track names which are present on spotify charts for maximum number of weeks.

Query 4 Result:

Sr. No.	Track Name	Weeks on Chart	Artist Genre
1	Mi Cuarto	36	dominican pop
2	5 Days	17	hands up
3	Otra Noche	12	nu-cumbia
4	One Night in Tokyo	5	finnish heavy metal
5	Scrivile Scemo	60	bergamo indie
6	Om du dro ifra mæ	4	trondersk musikk
7	Perlahan	54	dangdut
8	Mon Amour - Remix	19	latin talent show
9	Nothing Else Matters	1	chamber ensemble
10	Te Voy Amar	42	folklore paraguay

Query 4 Result:

Sr. No.	Track Name	Weeks on Chart	Artist Genre
11	Enemy	12	christian trap
12	Dragostea Ta	61	romanian rap
13	Paradise (feat. Dermot Kennec	40	pop house
14	Танцы	2	russian post-punk
15	Andai Aku Bisa	19	indonesian city pop
16	Nattely	6	danish indie pop
17	My Resort	26	japanese trap
18	u.	17	drift phonk
19	Tujh Mein Rab Dikhta Hai	6	classic bollywood
20	Ce N-as Da	1	muzica etno

Query 4 Result

REFERENCES

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